

STAT 4830: Modded ViT Training Pipeline

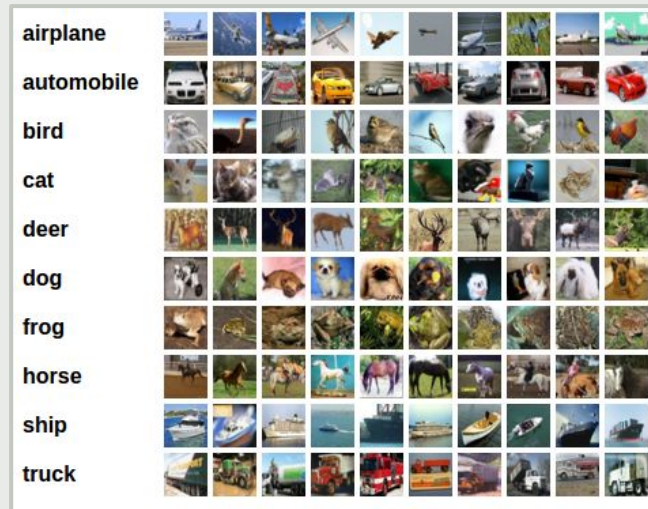
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Problem & Goal

- Build **reproducible ViT training pipeline** for CIFAR-10
 - Primary objective: **minimize wall-clock time to reach 94% test accuracy**
 - Secondary metrics:
 - Throughput (images/sec)
 - Peak GPU memory
 - Stability of train/test accuracy
 - Motivation: faster experimentation with architectures, hyperparameters, and optimizations
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Dataset & Model Choice

- Dataset: CIFAR-10 (50k train / 10k test, 10 classes)
- Two model paths:
 - **Pre-trained ViT (ImageNet)** → faster convergence
 - **Scratch ViT** → architecture experimentation
- For pre-trained ViT: resize images to **224×224**, ImageNet normalization
- Head replaced for 10-class classification



Mathematical Objective

- Minimize cross-entropy loss:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{p}_{i,c})$$

- ViT pipeline:
 - Patch embedding $\rightarrow$ tokens + [CLS]
 - Transformer encoder stack
 - [CLS] $\rightarrow$ linear head $\rightarrow$ logits (10 classes)
 - Optimization target: reach 94% accuracy as fast as possible
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Pipeline Architecture

- Modular PyTorch design
 - `model.py`: ViT / pretrained wrapper
 - `utils.py`: loaders, training, validation, logging
 - Deterministic seeding for reproducibility
 - CIFAR-10 loaders with augmentation
 - Early-stop capable training loop
 - Logs per epoch:
 - Train/Test accuracy
 - Throughput
 - Peak GPU memory
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Configuration & Sanity Checks

- Key settings:
 - Batch size: 64 (pretrained) / 128 (scratch)
 - LR = $3e-4$, epochs ≤ 100
 - Forward-pass sanity test $\rightarrow$ output shape (B,10)
 - Loss checked finite
 - Parameter count $\approx 5.5M$
 - Repeated runs with same seed $\rightarrow$ identical behavior
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Improvements & Next Steps

- Add **early stopping at 94%** for precise time-to-target
 - Enable **AMP (mixed precision)**
 - Try stronger augmentations: Mixup, RandAugment
 - Tune patch size, depth, heads for CIFAR-10
 - Add YAML/JSON configs for systematic experiments
 - Compare multiple ViT variants and smaller models
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